

1. O/N/23/22/Q6

The loudspeaker shown in Fig. 6.1 produces a sound of frequency 800 Hz.

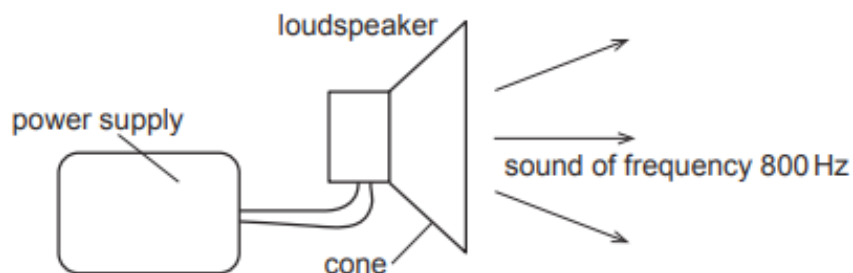


Fig. 6.1

- (a) State what is meant by 'frequency'.

.....
 [2]

- (b) Explain whether the sound produced by the speaker is audible to a human with healthy hearing.

.....
 [1]

- (c) Describe the motion of the cone in Fig. 6.1 and explain how this causes the production of sound.

.....

 [3]

- (d) The Sun is a very large and extremely violent ball of gas. Electromagnetic radiation from explosions on the surface of the Sun reaches the Earth.

Explain why sound from explosions on the surface of the Sun does **not** reach the Earth.

.....

 [2]

[Total: 8]

2. O/N/22/22/Q4

A sound wave, travelling in air or water, contains compressions and rarefactions.

- (a) Describe an experiment to show that a medium is required to transmit sound waves. You may draw a labelled diagram if you wish.

.....

.....

.....

.....

..... [3]

- (b) When a sound wave passes through air or water, the molecules of the air or water move.
Describe the motion of the molecules due to the sound wave.

.....

.....

..... [2]

- (c) Describe what is meant by 'a rarefaction'.

.....

..... [1]

(d) Fig. 4.1 shows a dolphin.

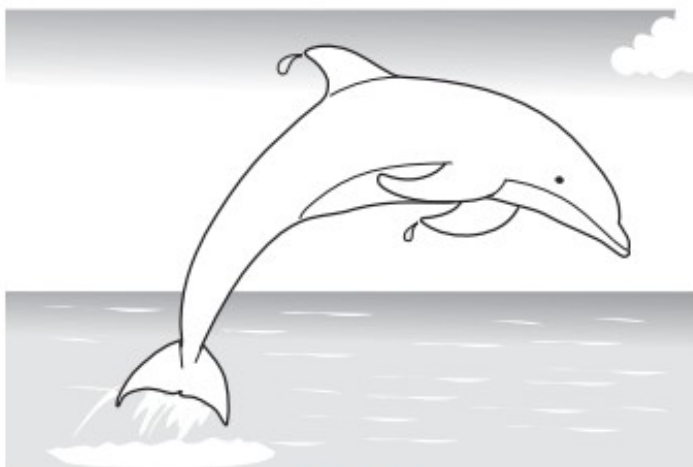


Fig. 4.1

The range of audible sound frequencies for the dolphin is 75 Hz to 140 000 Hz.

In water, the speed of sound is 1500 m/s.

Calculate the smallest wavelength of sound in water that the dolphin can hear.

wavelength = [2]

[Total: 8]

3. M/J/20/21/Q9

Ultrasound and X-rays are both used in medical imaging.

- (a) (i) Define what is meant by *ultrasound*.

.....
 [2]

- (ii) Describe what happens to ultrasound waves as they meet the boundary between two different materials.

.....
 [2]

- (iii) To produce the image of an unborn child, an ultrasound emitter and receiver are placed close together on the mother's skin.

Fig. 9.1 shows pulses detected by the receiver.

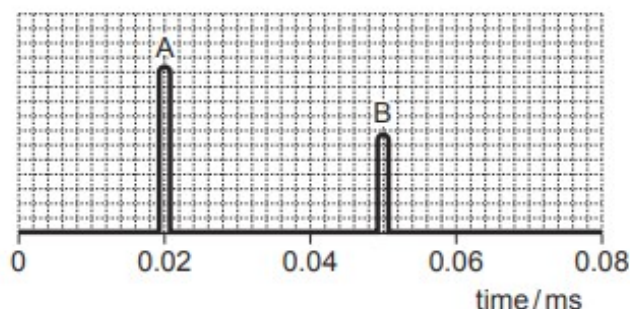


Fig. 9.1

Pulse A is the emitted pulse and pulse B is the first pulse that returns from the unborn child.

The average speed of ultrasound in human tissue is 1500 m/s.

Calculate the distance between the emitter and the child.

distance = [3]

- (iv) The speed of ultrasound in human tissue is close to the speed of sound in water.

Suggest approximate values for the speed of sound in gases and solids.

speed in gases

speed in solids

[2]

- (b) Fig. 9.2 shows an X-ray image of a hand. An X-ray detector is placed just below the hand. An image of the bones and human tissue around the bones is formed on a screen by the detector.

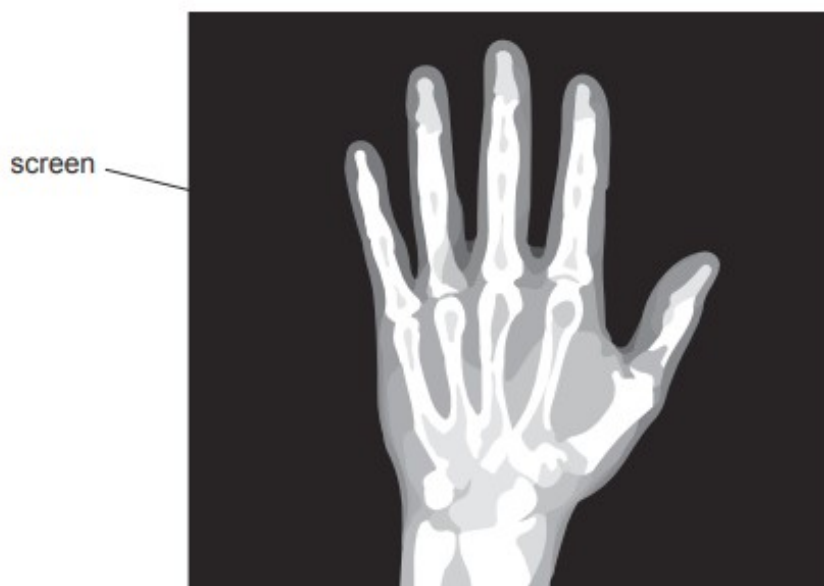


Fig. 9.2

- (i) Describe what happens to the X-rays to produce the image.

.....

.....

.....

.....

..... [3]

- (ii) The wavelength of the X-rays used is 2.0×10^{-9} m. The speed of electromagnetic waves is 3.0×10^8 m/s.

Calculate the frequency of the X-rays.

frequency = [2]

- (iii) Suggest one reason why X-rays are **not** used to form an image of an unborn child.

.....

..... [1]

4. O/N/19/22/Q5

Both sound and ultrasound are longitudinal waves.

(a) Describe what is meant by *longitudinal*.

.....
.....
..... [2]

(b) Fig. 5.1 shows an ultrasound cleaner being used to clean jewellery.

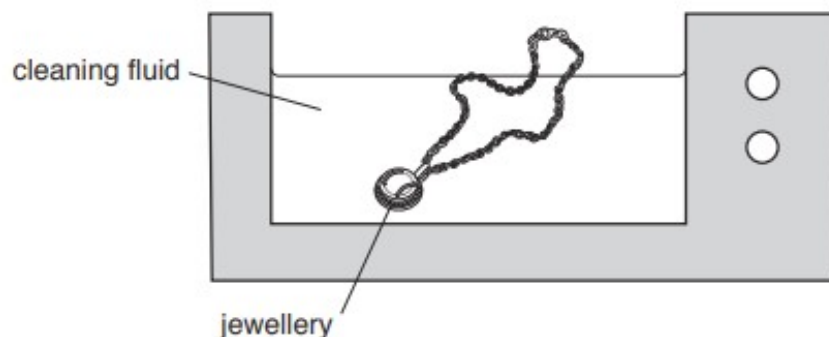


Fig. 5.1

The jewellery is lowered into the cleaning fluid and ultrasound waves of frequency 42 000 Hz are produced in the fluid.

(i) The speed of ultrasound in the fluid is 1500 m/s.

Calculate the wavelength of the ultrasound in the fluid.

wavelength = [2]

(ii) Describe how the ultrasound cleans the jewellery.

.....
.....
..... [2]

[Total: 6]

5. M/J/19/21/Q10(b)

A student reads in a textbook that the speed of sound in air is about 300 m/s.

He plans an experiment to check this value.

He asks another student to stand a long distance away in a large open space.

One student has a starting pistol. The pistol, when fired, produces a loud sound and a puff of smoke at the same instant.

- (i) Other pieces of apparatus are needed for this experiment in order to measure the speed of sound.

Name two other pieces of apparatus that are needed.

1.

2.

[2]

- (ii) Describe how all the measurements are made.

.....

.....

.....

.....

.....

.....

.....

..... [3]

- (iii) State typical values for the speed of sound in:

1. a liquid

.....

2. a solid.

.....

[2]

6. O/N/18/22/Q9

Both sound and ultrasound are waves that travel in solids, liquids and gases.

(a) State how sound differs from ultrasound.

.....
..... [1]

(b) Sound and ultrasound are longitudinal waves that consist of compressions and rarefactions.

(i) Explain what is meant by a wave and describe how a longitudinal wave differs from a transverse wave.

.....
.....
.....
.....
..... [3]

(ii) Fig. 9.1 represents a longitudinal wave of frequency 25 000 Hz travelling in oil.

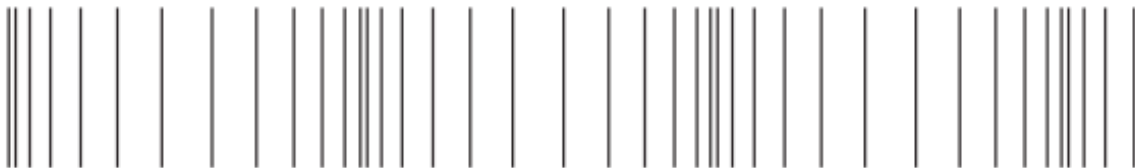


Fig. 9.1 (full scale)

1. On Fig. 9.1, mark two points at the centre of two different rarefactions and label each one R. [1]
2. On Fig. 9.1, draw a double-headed arrow to indicate a distance that is equal to one wavelength of the wave. [1]
3. Measure the length of the arrow drawn on Fig. 9.1 and use it to determine the speed of the wave in oil.

speed = [2]

(c) A sound wave travelling in a liquid, passes into air.

(i) State what happens to the speed of the sound wave as it enters the air.

..... [1]

- (ii) The wave in the liquid travels towards the surface at an angle. Fig. 9.2 shows the centres of the compressions of the sound wave in the liquid.

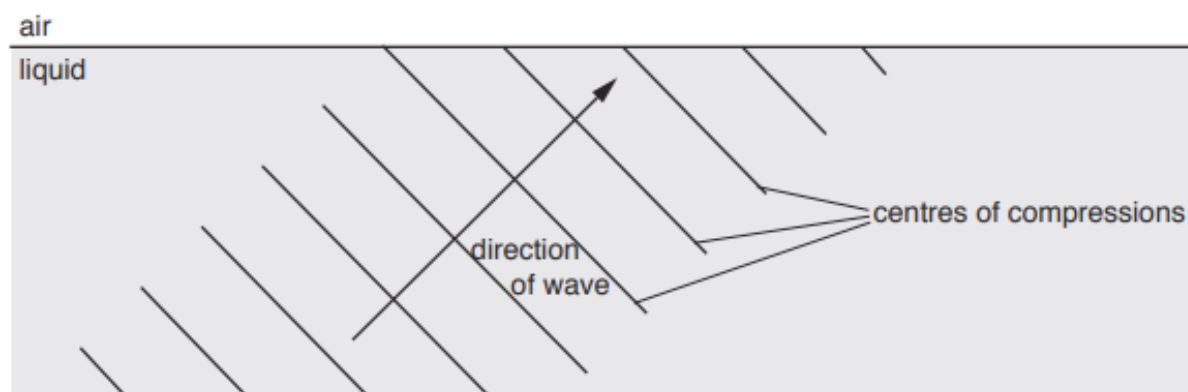


Fig. 9.2

Some compressions shown have reached the liquid-air boundary. The parts of these compressions in the air are not shown on Fig. 9.2.

On Fig. 9.2, draw the parts of these compressions that are in the air. [3]

- (d) Describe, in outline, how ultrasound is used in cleaning.

.....

.....

.....

.....

..... [3]

7. O/N/17/22/Q6

Fig. 6.1 shows a girl standing a few hundred metres in front of a large building.



Fig. 6.1 (not to scale)

The girl uses a signal generator and a loudspeaker to send a short pulse of sound towards the building. The sound has a frequency of 3700 Hz .

A short time later, the girl hears an echo.

- (a) State what is meant by the term *echo*.

.....
.....[1]

- (b) The pitch of the echo is the same as that of the original sound but the echo is not as loud.

State what has happened to

- (i) the amplitude of the sound wave,

.....[1]

- (ii) the frequency of the sound wave.

.....[1]

- (c) The speed of sound in air is 330 m/s .

Calculate the wavelength of this sound.

wavelength =[2]

8. O/N/17/21/Q5

Some medical processes involve the use of ultrasound.

(a) Explain what is meant by *ultrasound*.

.....

.....

.....

..... [2]

(b) Explain briefly how ultrasound is used in pre-natal scanning.

.....

.....

.....

.....

.....

..... [3]

9. M/J/17/22/Q6

- (a)** Explain the difference between a longitudinal wave and a transverse wave.

You may draw a diagram, if you wish, to help your explanation.

.....

.....

.....

.....

.....[2]

- (b)** A sound wave has a frequency of 3.8 kHz and a speed of 330 m/s.

- (i)** Calculate the wavelength of the sound.

wavelength =[2]

- (ii)** A different sound has a frequency of 3.8 Hz.

State and explain whether a human with normal hearing is able to hear this sound.

.....

.....

.....[1]

10. O/N/16/21/Q5

Fig. 5.1 shows a loudspeaker inside a glass bell-jar that contains air.

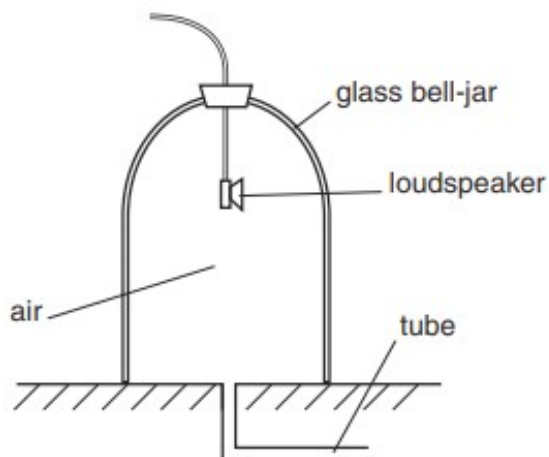


Fig. 5.1

The loudspeaker produces a sound of frequency 2200 Hz.

(a) State what is meant by the *frequency* of a sound.

.....

.....

.....[1]

(b) (i) The speed of sound in air is 330 m/s.

Calculate the wavelength of the sound in the air.

wavelength =[2]

(ii) The sound passes from the air into the glass.

State what happens to

1. the frequency of the sound,

.....

2. the speed of the sound.

.....

[1]

(c) (i) Describe

1. how sound is produced by the loudspeaker,

.....
.....
.....[1]

2. how sound is transmitted through the air.

.....
.....
.....[2]

(ii) A pump is connected to the tube at the bottom of the glass bell-jar. The pump is switched on and it removes the air from the glass bell-jar.

Explain why the volume of the sound heard outside the glass bell-jar decreases.

.....
.....
.....[1]

11. O/N/15/21/Q7

Audible sound waves and ultrasound waves both consist of compressions and rarefactions.

- (a)** Describe one difference between a compression and a rarefaction.

.....
.....
..... [1]

- (b)** In a particular medium, the speed of both audible sound and ultrasound waves is v .

- (i)** State the equation that relates the wavelength λ of a sound wave to its frequency f .

.....
..... [1]

- (ii)** Explain how, in this medium, the wavelength of an audible sound wave compares with the wavelength of an ultrasound wave.

.....
.....
.....
..... [1]

- (c)** Describe one use of ultrasound waves.

.....
.....
.....
.....
..... [2]

12. O/N/14/21/Q5

The device shown in Fig. 5.1 uses the reflection of ultrasound to measure distances.

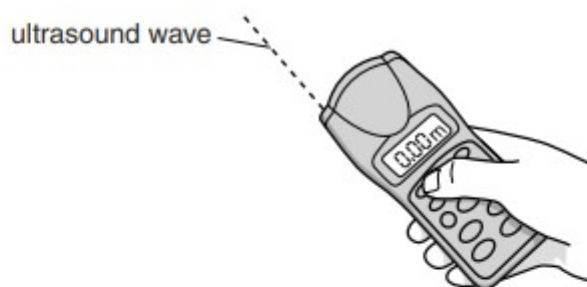


Fig. 5.1

- (a) State what is meant by *ultrasound*.

.....

.....

.....[2]

- (b) Fig. 5.2 shows a builder using the ultrasound device to measure the width of a room.

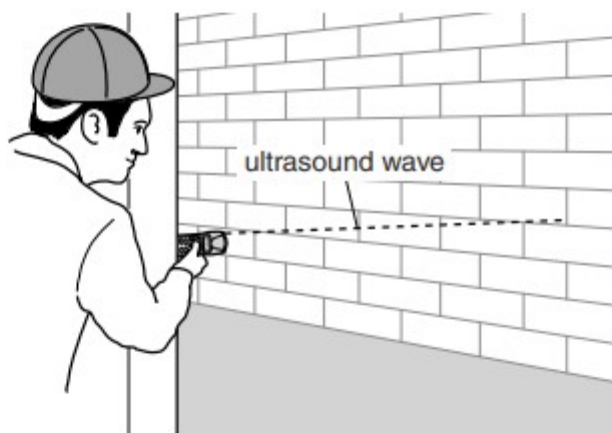


Fig. 5.2

The ultrasound device is placed against one wall and it emits an ultrasound wave that reflects back from the opposite wall.

The time between sending out the ultrasound wave and receiving the reflection is 0.030 s. The speed of ultrasound in air is 340 m/s.

Calculate the distance between the device and the opposite wall.

distance =[2]

13. M/J/14/21/Q10

A sound wave travels through air.

- (a) Explain what is meant by the *wavelength* of a wave.

.....
..... [1]

- (b) Fig. 10.1 represents air molecules in the sound wave at one instant.



Fig. 10.1

- (i) Describe the motion of a molecule of air due to the sound.

.....
..... [2]

- (ii) State one difference between the motion of molecule A and the motion of molecule B.

.....
..... [1]

- (c) Describe an experiment that shows that a medium is needed to transmit sound waves. Draw a labelled diagram of the apparatus.

.....
.....
.....
.....
.....
..... [4]

- (d) A short pulse of sound waves produces an echo from a wall 20 m away. The echo arrives back at the source of the sound 0.12 s after the pulse is produced.

The wavelength of the sound wave is 30 mm.

- (i) State what is meant by an *echo*.

.....
..... [1]

- (ii) Calculate the speed of the sound.

speed = [2]

- (iii) Calculate the frequency of the sound.

frequency = [3]

- (iv) The sound wave produces compressions and rarefactions in the air as it passes.

Calculate the distance between a compression and the nearest rarefaction.

distance = [1]

14. O/N/13/22/Q5

A physics textbook states that sound is a longitudinal pressure wave with a frequency within the audible range.

(a) Explain what is meant by a *longitudinal wave*.

.....
.....
.....[2]

(b) (i) State the approximate range of audible frequencies.

highest frequency:

lowest frequency:

[2]

(ii) The speed of sound in air is 330 m/s. Using your answer in (i), calculate the shortest wavelength in air of sound in the audible range.

wavelength =[2]

15. M/J/11/21/Q6

Fig. 6.1 shows a bat.



Fig. 6.1

Bats emit short bursts of ultrasound. The echoes of the ultrasound help the bat find insects and prevent the bat flying into objects.

(a) State what is meant by an *echo*.

.....
..... [1]

(b) Fig. 6.2 shows the variation with time of air pressure caused by a burst of ultrasound.



Fig. 6.2

On Fig. 6.2, draw a possible echo formed by this burst of ultrasound. [2]

(c) State

(i) the approximate range of frequencies of sound audible to humans,

..... [1]

(ii) how the frequency of ultrasound differs from frequencies in this audible range.

.....
..... [1]

16. M/J/10/22/Q10

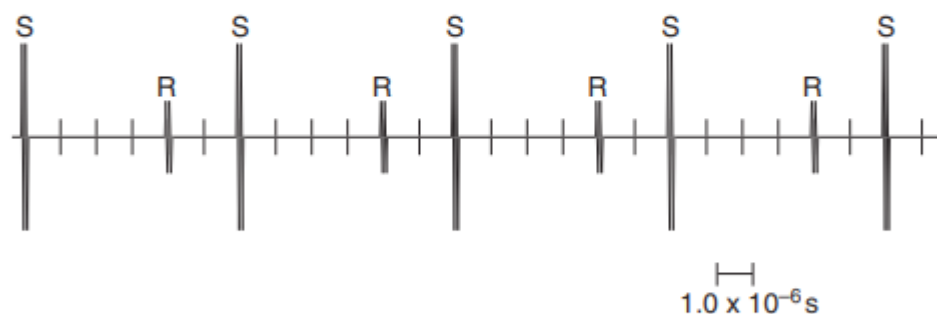
- (a) Describe a method for measuring the speed of sound in air. In your account, state clearly
- how the sound is made,
 - what measurements are taken,
 - how the result is calculated,
 - one precaution to produce an accurate result.

[5]

- (b)** Ultrasound is used in quality control to detect cracks in metal.

Pulses of ultrasound are sent into the metal from a transmitter. A detector is placed next to the transmitter on the front surface of the metal.

Fig. 10.1 shows the oscilloscope trace of the ultrasound pulses produced if the metal contains no cracks.

**Fig. 10.1**

One division along the x-axis represents 1.0×10^{-6} s.

Pulses labelled S are the pulses initially sent out from the transmitter. Each pulse labelled R is the reflection from the back surface of the metal of the previous pulse S.

- (i) State what is meant by *ultrasound*.

.....
.....[2]

- (ii) Use Fig. 10.1 to calculate the number of pulses sent out by the source in one second.

number =[2]

- (iii) Suggest two reasons why the amplitude of R is less than the amplitude of S.

1.
.....
2.
.....
[2]

- (iv) Some time later, the piece of metal is tested again. It now has a small crack half-way between the front surface and the back surface.

On Fig. 10.1, draw the position and size of the pulses produced by this crack.

Label each of these pulses C. [2]

- (v) A second beam of ultrasound has a frequency of $8.0 \times 10^6 \text{ Hz}$ and a speed of 4000 m/s in the metal.

Calculate the wavelength of this ultrasound in the metal.

wavelength =[2]

17. O/N/09/Q5

Fig. 5.1 is a full-scale diagram that represents a sound wave of frequency 5.1 kHz in air.

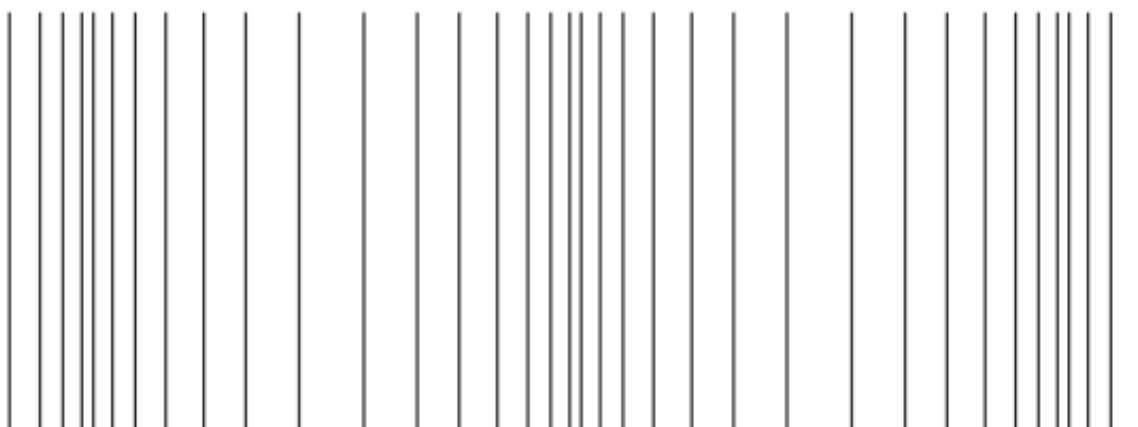


Fig. 5.1

- (a) (i) On Fig. 5.1, mark the position of one compression with the letter C and of one rarefaction with the letter R. [2]

- (ii) Using the full-scale diagram, measure the wavelength of this sound wave.

wavelength = [1]

- (iii) Calculate the speed of sound in air.

speed = [3]

- (b) A sound wave is longitudinal. Describe how a longitudinal wave differs from a transverse wave.

.....

 [2]

18. M/J/09/Q6

Fig. 6.1 shows how ultrasound is used to produce an image of the heart.

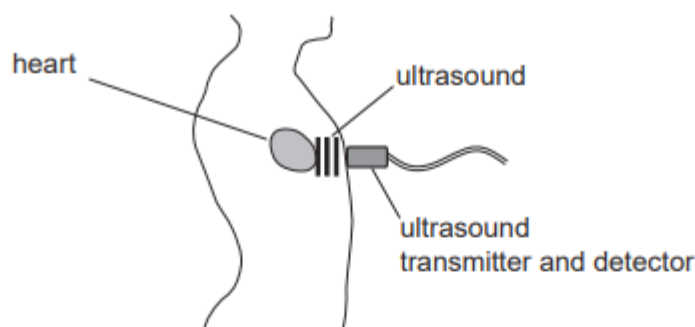


Fig. 6.1

- (a) Define *ultrasound*.

.....
 [1]

- (b) The ultrasound has a wavelength of 1.2×10^{-3} m. The speed of the ultrasound in the human body is 1500 m/s.

Calculate the frequency of the ultrasound.

frequency = [2]

- (c) Ultrasound is a longitudinal wave.
 Describe how particles in the body move as the ultrasound passes.

You may draw a diagram if you wish.

.....

 [2]

- (d) There are small bubbles of gas in the body.

Explain why these bubbles expand and contract as the ultrasound passes.

.....

 [1]

19. O/N/07/Q6

Fig. 6.1 shows the cone of a loudspeaker.

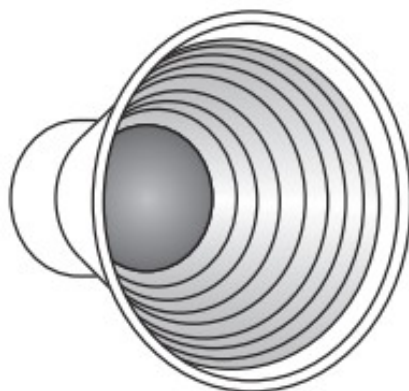


Fig. 6.1

- (a) Sound is being produced. Describe in detail the behaviour of the cone and the air near to it.

.....

.....

.....

.....

..... [2]

- (b) The lowest frequency that a human can hear is 20 Hz.

- (i) State the highest frequency that a human with normal hearing can hear.

..... [1]

- (ii) Calculate the longest wavelength of sound that a human can hear. The speed of sound in air is 340 m/s.

wavelength = [2]

20. M/J/04/Q4

Fig. 4.1 shows a student clapping in front of a vertical wall. The wall reflects the sound.

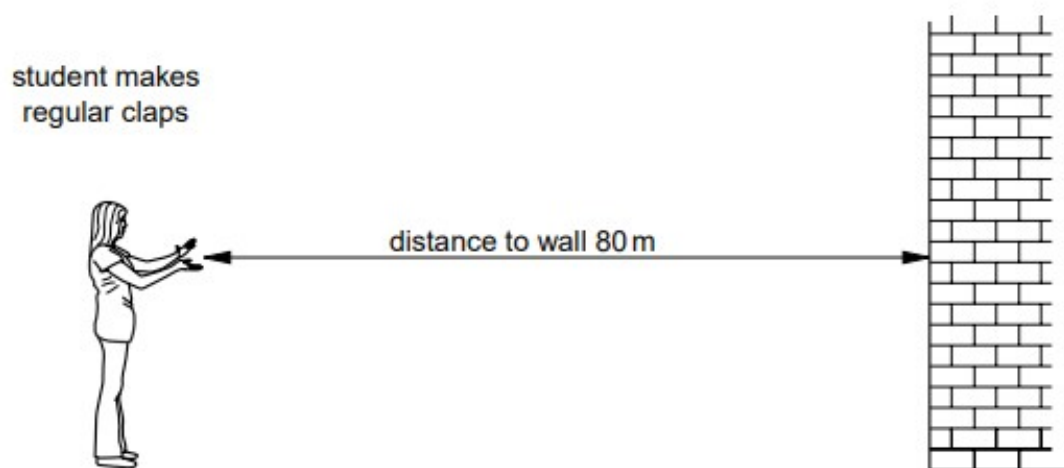


Fig. 4.1

The student changes the number of claps made in 1 minute until the reflection of each clap returns to her at exactly the same time as she makes the next clap.

The speed of sound in air is 330 m/s.

(a) Explain what is meant by *speed*.

.....
[1]

(b) Calculate the time between claps.

time = [3]

(c) Calculate the number of claps in 1 minute.

number of claps = [2]